

Number: Place Value								
	Nursery (Mathematics)	Reception (Mathematics)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can develop fast recognition of up to 3 objects, without having to count them individually (subitising). I can say one number for each item in order: 1, 2, 3, 4, 5. I know that the last number reached when counting a small set of objects tells you how many there are in total. I can show 'finger numbers' up to 5. I can recite numbers past 5. I can link numerals and amounts: for example, showing the right number of objects to match the numeral up to	I can begin to develop a sense of the number system by verbally counting forward to and beyond 20, pausing at each multiple of 10. I can play games that involve moving along a numbered track, and understand that larger numbers are further along the track. I can explore and represent patterns within numbers up to 10, including evens and odds, and double facts.	I can count within 100, forwards and backwards, starting with any number. I can reason about the location of numbers to 20 within the linear number system, including comparing using $<$, $>$ and $=$. I can read and write numbers to 100 in numerals. I can read and write numbers 1-20 in numerals and words.	I can recognise the place value of each digit in two-digit numbers, and compose and decompose two-digit numbers using standard and nonstandard partitioning. I can reason about the location of any two-digit number in the linear number system, including identifying the previous and next multiple of 10. I can identify, represent and estimate numbers using different representations, including the number line. I can read and write numbers to 100 in numerals and	I can understand that 10 tens are equivalent to 1 hundred, and that 100 is 10 times the size of 10; apply this to identify and work out how many 10s there are in other three-digit multiples of 10. I can recognise the place value of each digit in three-digit numbers, and compose and decompose three-digit numbers using standard and non-standard partitioning. I can reason about the location of any three-digit number in the linear number system, including identifying the previous and next multiple of 100 and	I can understand that 10 hundreds are equivalent to 1 thousand, and that 1,000 is 10 times the size of 100; apply this to identify and work out how many 100s there are in other four-digit multiples of 100. I can recognise the place value of each digit in four-digit numbers, and compose and decompose four-digit numbers using standard and nonstandard partitioning. I can reason about the location of any four-digit number in the linear number system, including identifying the previous and next multiple of 1,000	I can understand that 10 tenths are equivalent to 1 one, and that 1 is 10 times the size of 0.1; know that 100 hundredths are equivalent to 1 one, and that 1 is 100 times the size of 0.01; know that 10 hundredths are equivalent to 1 tenth, and that 0.1 is 10 times the size of 0.01. I can recognise the place value of each digit in numbers up to 1 million, with up to 3 decimal places, and compose and decompose numbers with up to 3 decimal places using standard and nonstandard partitioning. I can reason about	I can understand the relationship between powers of 10 from 1 hundredth to 10 million, and use this to make a given number 10, 100, 1,000, 1 tenth, 1 hundredth or 1 thousandth times the size (multiply and divide by 10, 100 and 1,000). I can recognise the place value of each digit in numbers up to 10 million, including decimal fractions, and compose and decompose numbers up to 10 million using standard and nonstandard partitioning. I can reason about the location of any

	5. I can compare quantities using language ‘more than,’ ‘fewer than.’ I can explore composition of numbers to 10. I can solve real-world mathematical problems with numbers up to 5.			words. I can count forwards and backwards in multiples of 2, 3 and 5 from 0, and in tens from any number.	10. I can divide 100 into 2, 4, 5 and 10 equal parts, and read scales/number lines marked in multiples of 100 with 2, 4, 5 and 10 equal parts. I can read and write numbers up to 1000 in numerals and words. I can count forwards and backwards in multiples of 4, 8 and 50 from 0, and in hundreds from any number.	and 100, and rounding to the nearest of each. I can divide 1,000 into 2, 4, 5 and 10 equal parts, and read scales/number lines marked in multiples of 1,000 with 2, 4, 5 and 10 equal parts. I can count backwards through zero to include negative numbers. I can read Roman numerals to 100 (I to C) and know that over time, the numeral system changed to include the concept of zero and place value.	the location of any number with up to 3 decimal places in the linear number system, including identifying the previous and next multiple of powers of 10, 1 and 0.1 and rounding to the nearest of each. I can divide 1 into 2, 4, 5 and 10 equal parts, and read scales/number lines marked in units of 1 with 2, 4, 5 and 10 equal parts. I can count forwards or backwards in steps of powers of 10 for any given number up to 1 million. I can interpret negative numbers in context, count forwards and backwards with positive and negative whole numbers, including through zero. I can read Roman numerals to 1000 (M) and recognise years written in Roman numerals.	number up to 10 million, including decimal fractions, in the linear number system, and round numbers, as appropriate, including in contexts. I can divide powers of 10, from 1 hundredth to 10 million, into 2, 4, 5 and 10 equal parts, and read scales/number lines with labelled intervals divided into 2, 4, 5 and 10 equal parts. I can use negative numbers in context, and calculate intervals across zero.
Vocabulary	Numbers, total, compare, count, more than, fewer than	Concrete objects, count, digit, equal, even number, number line, number sentence,	Compare, concrete objects, consecutive, count, diagram, digit,	Compare, concrete objects, consecutive, count, diagram, digit,	Compare, consecutive, digit, equal, even number, integer, number	Compare, consecutive, digit, equal, even number, integer, negative	Compare, consecutive, digit, equal, even number, integer, negative	Compare, consecutive, digit, equal, even number, integer, negative

		number track, odd number, pattern, quantity, sequence	equal, even number, hundred square, number line, number sentence, number track, numeral, odd number, partition, pattern, quantity, sequence, sign	equal, even number, hundred square, number line, number sentence, number track, numeral, odd number, partition, pattern, quantity, sequence, sign	line, numeral, odd number, partition, pattern, sequence, sign	number, number line, numeral, odd number, partition, pattern, positive number, Roman numerals, round, sequence, sign	number, number line, numeral, odd number, partition, pattern, positive number, Roman numerals, round, sequence, sign	number, number line, numeral, odd number, partition, pattern, positive number, round, sequence, sign
--	--	---	---	---	--	--	--	---

Number: Addition & Subtraction

	Nursery (Mathematics)	Reception (Mathematics)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can explore how numbers can be made in different ways using objects and practical experiences. I can recognise that a group of objects can be separated into different parts and still have the same total. I can combine two small groups of objects and find how many there are altogether. I can understand that the last number I say when counting tells me how many objects there are altogether. I can subitise up to 3 objects without counting them individually. I can show numbers to 5 using my fingers. I can solve simple real-world mathematical	I can automatically recall number bonds for numbers 0–5 and some number bonds to 10. I can explore the composition of numbers to 10 and understand that numbers can be made from different parts. I can partition and combine numbers within 10. I can subitise quantities up to 5. I can automatically show a given number to 5 using my fingers. I can use objects, pictures and mathematical representations to explore addition and subtraction. I can devise and record number stories using pictures, numbers and symbols.	I can develop my fluency in addition and subtraction facts within 10. I can compose numbers to 10 from 2 parts, and partition numbers to 10 into parts, including recognising odd and even numbers. I can read, write and interpret equations containing addition (+), subtraction (-) and equals (=) symbols, and relate additive expressions and equations to real-life contexts. I can add and subtract one-digit and two-digit numbers to 20.	I can secure my fluency in addition and subtraction facts within 20, through continued practice. I can recognise the subtraction structure of 'difference' and answer questions of the form, "How many more...?". I can add and subtract within 100 by applying related one-digit addition and subtraction facts: add and subtract only ones or only tens to/from a two-digit number; add and subtract any 2 two-digit numbers. I can show that addition of two numbers can be done in any order (commutative) and subtraction of one number from another cannot. I can recognise and use the inverse	I can secure my fluency in addition and subtraction facts that bridge 10, through continued practice. I can calculate complements to 100. I can add and subtract numbers mentally, including a three-digit number and a discrete place value, e.g. $273 + 40$. I can add and subtract up to three-digit numbers using columnar methods. I can manipulate the additive relationship: understand the inverse relationship between addition and subtraction, and how both relate to the part-part-whole structure; understand and use the commutative property of addition, and	I can add and subtract numbers with up to 4 digits using the formal written methods of columnar addition and subtraction where appropriate. I can solve addition and subtraction two-step problems in contexts, deciding which operations and methods to use and why.	I can add and subtract whole numbers with more than 4 digits, including using formal written methods (columnar addition and subtraction). I can add and subtract numbers mentally with increasingly large numbers. I can use rounding to check answers to calculations and determine, in the context of a problem, levels of accuracy. I can solve addition and subtraction multi-step problems in contexts, deciding which operations and methods to use and why.	I can understand that 2 numbers can be related additively, and quantify additive relationships. I can use a given additive calculation to derive or complete a related calculation, using arithmetic properties, inverse relationships, and place-value understanding.

	problems involving quantities up to 5.	I can solve practical addition and subtraction problems within 10.		relationship between addition and subtraction and use this to check calculations and solve missing number problems.	understand the related property for subtraction. I can estimate the answer to a calculation and use inverse operations to check answers.			
Vocabulary	Number bond	Add, subtract, altogether, part, whole, total, more, fewer, less, equal, number bond, number story	Addend, addition, compensation, decomposition, difference, facts, minus, number bond, plus, subtract, subtraction, subtrahend, sum, take away, total	Addend, addition, commutative, compensation, decomposition, difference, facts, minus, number bond, plus, subtract, subtraction, subtrahend, sum, take away, total	Addend, addition, approximation, columnar addition/subtraction, commutative, compensation, complement, decomposition, difference, estimate, exchange, facts, minus, plus, subtract, subtraction, subtrahend, sum, take away, total	Addend, addition, approximation, columnar addition/subtraction, compensation, complement, decomposition, difference, estimate, exchange, facts, minus, plus, subtract, subtraction, subtrahend, sum, take away, total	Addend, addition, approximation, columnar addition/subtraction, compensation, complement, decomposition, difference, estimate, exchange, facts, minus, plus, subtract, subtraction, subtrahend, sum, take away, total	Addend, addition, approximation, compensation, complement, decomposition, difference, estimate, exchange, facts, minus, plus, subtract, subtraction, subtrahend, sum, take away, total

Number: Multiplication & Division

	Nursery (Mathematics)	Reception (Mathematics)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can share objects between people or groups through practical play. I can recognise when objects have been shared fairly or unfairly. I can explore making groups containing the same number of objects. I can use language such as same, more and fewer when comparing groups. I can explore doubles using objects and practical experiences.	I can share quantities fairly between groups and recognise when each group has an equal amount. I can group objects so that each group contains an equal number. I can explore and represent doubles within 10. I can recognise and recall some double facts. I can explore how quantities can be shared and grouped in different ways. I can use objects, pictures and practical contexts to solve simple problems involving equal groups and sharing.	I can count forwards and backwards in multiples of 2, 5 and 10, up to 10 multiples, beginning with any multiple, and count forwards and backwards through the odd numbers. I can solve one-step multiplication and division problems by using concrete objects, pictorial representations and arrays, with support.	I can recognise repeated addition contexts, representing them with multiplication equations and calculating the product, within the 2, 5 and 10 multiplication tables. I can relate grouping problems where the number of groups is unknown to multiplication equations with a missing factor, and to division equations (quotitive division). I can calculate mathematical statements for multiplication and division within the multiplication tables and write them using the multiplication (X), division (÷) and equals (=) symbols. I can show that multiplication of two numbers can be done in any order	I can recall multiplication facts, and corresponding division facts, in the 10, 5, 2, 3, 4 and 8 multiplication tables, and recognise products in these multiplication tables as multiples of the corresponding number. I can apply place-value knowledge to known additive and multiplicative number facts (scaling facts by 10). I can apply known multiplication and division facts to solve contextual problems with different structures, including quotitive and partitive division.	I can recall multiplication and division facts up to 12X12, and recognise products in multiplication tables as multiples of the corresponding number. I can multiply two-digit and three-digit numbers by a one-digit number using formal written layout. I can solve division problems, with two-digit dividends and one-digit divisors, that involve remainders, and interpret remainders appropriately according to the context. I can apply place-value knowledge to known multiplicative number facts (scaling facts by 100) to multiply and	I can secure fluency in multiplication table facts, and corresponding division facts, through continued practice. I can apply place-value knowledge to known additive and multiplicative number facts (scaling facts by 1 tenth or 1 hundredth). I can multiply and divide numbers by 10, 100 and 1000; understand this as equivalent to making a number 10, 100 or 1000 times the size, or 1 tenth, 1 hundredth or 1 thousandth times the size. I can find factors and multiples of positive whole numbers, including common factors and common multiples, and express a given number as a product of 2 or 3	I can understand that 2 numbers can be related multiplicatively, and quantify multiplicative relationships (restricted to multiplication by a whole number). I can use a given multiplicative calculation to derive or complete a related calculation, using arithmetic properties, inverse relationships, and place-value understanding. I can divide numbers up to 4 digits by a two-digit whole number using the formal written method of long division, and interpret remainders as whole number remainders, fractions, or by rounding, as appropriate for the context. I can identify

				(commutative) and division of one number from another cannot.		divide mentally, including: multiplying by 0 and 1; dividing by 1; multiplying together three numbers. I can multiply and divide whole numbers by 10 and 100 (keeping to whole number quotients); understand this as equivalent to making a number 10 or 100 times the size. I can manipulate multiplication and division equations, and understand and apply the commutative property of multiplication. I can understand and apply the distributive property of multiplication. I can count in multiples of 6, 7, 9, 25 and 1000.	factors. I can multiply any whole number with up to 4 digits by any one- or two-digit number using a formal written method, including long multiplication for two-digit numbers. I can divide a number with up to 4 digits by a one-digit number using a formal written method, and interpret remainders appropriately for the context. I can know and use the vocabulary of prime numbers, prime factors and composite (non-prime) numbers. I can establish whether a number up to 100 is prime and recall prime numbers up to 19. I can recognise and use square numbers and cube numbers, and the notation for squared (2) and cubed (3).	common factors, common multiples and prime numbers. I can use knowledge of the order of operations to carry out calculations involving the four operations.
Vocabulary	Double, share	Share, group, equal, equally, same,	Array, divide, dividend, division,	Array, commutative, divide, dividend,	Array, divide, dividend, divisibility,	Array, commutative, distributive, divide,	Array, common factor, common	Array, brackets, common factor,

		double, twice, pair	divisor, double, multiplicand, multiplication, multiply, product, repeated addition, repeated subtraction, share	division, divisor, double, multiplicand, multiplication, multiplication table, multiply, product, repeated addition, repeated subtraction, share	division, divisor, double, multiplicand, multiplication, multiplication table, multiply, product, quotient	dividend, divisibility, division, divisor, double, factor, multiple, multiplicand, multiplication, multiplication table, multiply, placeholder, product, quotient, remainder, short multiplication	multiple, cube number, divide, dividend, divisibility, division, divisor, double, factor, factorise, long multiplication, multiple, multiplicand, multiplication, multiplication table, multiply, placeholder, prime factor, prime number, product, quotient, remainder, short division, short multiplication, square number	common multiple, divide, dividend, divisibility, division, divisor, double, factor, factorise, long division, long multiplication, multiple, multiplicand, multiplication, multiplication table, multiply, order of operation, placeholder, prime factor, prime number, product, quotient, remainder, short division, short multiplication
--	--	---------------------	--	--	--	--	--	--

Number: Fractions, Decimals & Percentages

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can recognise, find and name halves and quarters of objects, shapes or quantities.	I can recognise, find, name and write $\frac{1}{3}$, $\frac{1}{4}$, $\frac{2}{4}$ and $\frac{3}{4}$ of a length, shape, set of objects or quantity. I can write simple fractions, e.g. $\frac{1}{2}$ of 6 = 3, and recognise the equivalence of $\frac{1}{2}$ and $\frac{2}{4}$.	I can interpret and write proper fractions to represent 1 or several parts of a whole that is divided into equal parts. I can find unit fractions of quantities using known division facts (multiplication tables fluency). I can reason about the location of any fraction within 1 in the linear number system. I can add and subtract fractions with the same denominator, within 1. I can count up and down in tenths; recognise that tenths arise from dividing an object into 10 equal parts and in dividing one-digit numbers or quantities by 10. I can recognise and show, using diagrams, equivalent fractions with small denominators.	I can reason about the location of mixed numbers in the linear number system. I can convert mixed numbers to improper fractions and vice versa. I can add and subtract improper and mixed fractions with the same denominator, including bridging whole numbers. I can recognise and show, using diagrams, families of common equivalent fractions. I can count up and down in hundredths; recognise that hundredths arise when dividing an object by one hundred and dividing tenths by ten. I can recognise and write decimal equivalents of any number of tenths, hundredths, and equivalents of $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{3}{4}$. I can find the effect of dividing a one- or two-digit number by 10 and 100, identifying the value of the digits in the answer as ones, tenths and hundredths.	I can find non-unit fractions of quantities. I can find equivalent fractions and understand that they have the same value and the same position in the linear number system. I can recall decimal fraction equivalents for $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$ and $\frac{1}{10}$, and for multiples of these proper fractions. I can compare and order fractions whose denominators are all multiples of the same number. I can recognise mixed numbers and improper fractions and convert from one form to the other and write mathematical statements > 1 as a mixed number [for example, $\frac{6}{5} = 1 \frac{1}{5}$]. I can add and subtract fractions with the same denominator and denominators that are multiples of the same number. I can multiply proper fractions and mixed numbers by whole numbers, supported by materials and diagrams. I can read and write decimal numbers as fractions [for example, $0.71 = \frac{71}{100}$]. I can recognise and use thousandths and relate them to	I can recognise when fractions can be simplified, and use common factors to simplify fractions. I can express fractions in a common denominator and use this to compare fractions that are similar in value. I can compare fractions with different denominators, including fractions greater than 1, using reasoning, and choose between reasoning and common denominator as a comparison strategy. I can multiply simple pairs of proper fractions, writing the answer in its simplest form [for example, $\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$]. I can divide proper fractions by whole numbers [for example, $\frac{1}{3} \div 2 = \frac{1}{6}$]. I can associate a fraction with division and calculate decimal fraction equivalents [for example, 0.375] for a simple fraction [for example, $\frac{3}{8}$]. I can multiply one-digit numbers with up to two decimal places by whole numbers.

				I can round decimals with one decimal place to the nearest whole number. I can compare numbers with the same number of decimal places up to two decimal places. I can solve simple measure and money problems involving fractions and decimals to two decimal places.	tenths, hundredths and decimal equivalents. I can recognise the per cent symbol (%) and understand that per cent relates to 'number of parts per hundred', and write percentages as a fraction with denominator 100, and as a decimal. I can solve problems which require knowing percentage and decimal equivalents of $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$, $\frac{1}{5}$, $\frac{2}{5}$, $\frac{3}{5}$, and those fractions with a denominator of a multiple of 10 or 25.	
Vocabulary	Fraction	Equivalent fractions, fraction	Decimal, decimal fraction, denominator, equivalent fractions, fraction, numerator, unit fraction	Decimal, decimal fraction, denominator, equivalent fractions, fraction, improper fraction, mixed number, numerator, unit fraction	Decimal, decimal fraction, denominator, equivalent fractions, fraction, improper fraction, mixed number, numerator, percentage, unit fraction	Decimal, decimal fraction, denominator, equivalent fractions, fraction, improper fraction, mixed number, numerator, percentage, recurring decimal, simplify, unit fraction

Number: Ratio & Algebra	
	Year 6
Substantive knowledge & Skills	I can solve problems involving ratio relationships. I can solve problems involving the calculation of percentages [for example, of measures, and such as 15% of 360] and the use of percentages for comparison. I can solve problems involving similar shapes where the scale factor is known or can be found. I can solve problems involving unequal sharing and grouping using knowledge of fractions and multiples. I can use simple formulae. I can generate and describe linear number sequences. I can express missing number problems algebraically. I can solve problems with 2 unknowns. I can enumerate possibilities of combinations of two variables.
Vocabulary	Algebra, expression, formula, proportion, ratio, ratio notation, rule, scale factor

Measurement								
	Nursery (Mathematics)	Reception (Mathematics)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can make comparisons between objects relating to size, length, weight and capacity.	I can use measuring tools, such as balance scales, to make comparisons between objects relating to size, length, height, mass/weight and capacity. I can talk about, order and sequence time.	I can compare, describe and solve practical problems for: lengths and heights, mass/weight, capacity and volume, and time. I can measure and begin to record lengths and heights, mass/weight, capacity and volume, and time (hours, minutes and seconds). I can recognise and know the value of different denominations of coins and notes. I can sequence events in chronological order. I can recognise and use language relating to dates, including days of the week, weeks, months and years. I can tell the time to the hour and half past the hour, and draw the hands on a	I can choose and use appropriate standard units to estimate and measure length/height in any direction (m/cm); mass (kg/g); temperature ($^{\circ}\text{C}$); capacity (litres/ml) to the nearest appropriate unit, using rulers, scales, thermometers and measuring vessels. I can compare and order lengths, mass, and volume/capacity and record the results using $<$, $>$ and $=$. I can recognise and use symbols for pounds (£) and pence (p) and combine amounts to make a particular value. I can find different combinations of coins that equal the same amount of money. I can solve simple	I can measure, compare, add and subtract: lengths (m/cm/mm); mass (kg/g); volume/capacity (l/ml). I can measure the perimeter of simple 2-D shapes. I can add and subtract amounts of money to give change, using both £ and p in practical contexts. I can tell and write the time from an analogue clock, including using Roman numerals from I to XII, and 12-hour and 24-hour clocks. I can estimate and read time with increasing accuracy to the nearest minute; record and compare time in terms of seconds, minutes and hours; use vocabulary such as o'clock, a.m./p.m., morning,	I can convert between different units of measure [for example, kilometre to metre; hour to minute]. I can measure and calculate the perimeter of a rectilinear figure (including squares) in centimetres and metres. I can find the area of rectilinear shapes by counting squares. I can estimate, compare and calculate different measures, including money in pounds and pence. I can read, write and convert time between analogue and digital 12- and 24-hour clocks. I can solve problems involving converting from hours to minutes; minutes to seconds; years to months; weeks to	I can convert between different units of metric measure (for example, kilometre and metre; centimetre and metre; centimetre and millimetre; gram and kilogram; litre and millilitre). I can understand and use approximate equivalences between metric units and common imperial units such as inches, pounds and pints. I can measure and calculate the perimeter of composite rectilinear shapes in centimetres and metres. I can calculate and compare the area of rectangles (including squares), and including using standard units, square centimetres (cm^2) and square metres (m^2) and	I can solve problems involving the calculation and conversion of units of measure, using decimal notation up to three decimal places where appropriate. I can use, read, write and convert between standard units, converting measurements of length, mass, volume and time from a smaller unit of measure to a larger unit, and vice versa, using decimal notation to up to three decimal places. I can convert between miles and kilometres. I can recognise that shapes with the same areas can have different perimeters and vice versa. I can recognise when it is possible to use formulae for

			clock face to show these times.	problems in practical contexts, including giving change. I can compare and sequence intervals of time. I can tell and write the time to five minutes, including quarter past/to the hour, and draw the hands on a clock face to show these times. I can know the number of minutes in an hour and the number of hours in a day.	afternoon, noon and midnight. I can know the number of seconds in a minute and the number of days in each month, year and leap year. I can compare durations of events [for example to calculate the time taken by particular events or tasks]..	days.	estimate the area of irregular shapes. I can estimate volume [for example, using 1 cm³ blocks to build cuboids (including cubes)] and capacity [for example, using water]. I can solve problems involving converting between units of time. I can use all four operations to solve problems involving measure [for example, length, mass, volume, money] using decimal notation, including scaling.	area and volume of shapes. I can calculate the area of parallelograms and triangles. I can calculate, estimate and compare volume of cubes and cuboids using standard units, including cubic centimetres (cm³) and cubic metres (m³), and extending to other units [for example, mm³ and km³].
Vocabulary	Tall, long, short, lighter, heavier, full, empty, nearly full, nearly empty	Balance, mass, heavier, lighter, more, less, most, least, long, short, longest, shortest, tall, tallest, height, minute, tomorrow, yesterday, first, then, after	Analogue clock, anticlockwise, capacity, chronological, clockwise, compare, digital clock, hour, length, mass, measure, minute, scale, time, unit, weight	Analogue clock, anticlockwise, capacity, centilitre, centimetre, clockwise, compare, denomination, digital clock, gram, hour, kilogram, length, litre, mass, measure, metre, millilitre, minute, pound sterling, scale, temperature, time, unit, weight	Analogue clock, anticlockwise, capacity, centilitre, centimetre, clockwise, compare, denomination, digital clock, gram, hour, kilogram, length, litre, mass, measure, metre, millilitre, millimetre, minute, perimeter, pound sterling, scale, second, time, unit, volume	Analogue clock, anticlockwise, area, centilitre, centimetre, clockwise, compare, convert, denomination, digital clock, gram, hour, kilometre, length, litre, mass, measure, metre, millilitre, millimetre, minute, perimeter, pound sterling, scale, second, square centimetre, square metre, square millimetre, time, unit, volume	Area, centilitre, centimetre, compare, convert, cubic centimetre, cubic metre, foot, gallon, gram, imperial unit, inch, kilogram, kilometre, length, litre, mass, measure, metre, metric unit, millilitre, millimetre, minute, ounce, perimeter, pint, pound (mass), scale, second, square centimetre, square metre, square millimetre, time, unit, volume, yard	Area, compare, convert, cubic centimetre, cubic metre, foot, gallon, gram, imperial unit, inch, kilogram, kilometre, length, litre, mass, measure, metre, metric unit, mile, millilitre, millimetre, ounce, perimeter, pint, pound (mass), scale, square centimetre, square metre, square millimetre, unit, volume, yard

Geometry: Shape

	Nursery (Mathematics)	Reception (Mathematics)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can talk about and explore 2D and 3D shapes (for example, circles, rectangles, triangles and cuboids) using informal and mathematical language: 'sides,' 'corners,' 'straight,' 'flat,' 'round.'	I can see, explore and discuss models of common 2D and 3D shapes with varied dimensions and presented in different orientations (for example, triangles not always presented on their base).	I can recognise common 2D and 3D shapes presented in different orientations, and know that rectangles, triangles, cuboids and pyramids are not always similar to one another.	I can use precise language to describe the properties of 2D and 3D shapes, and compare shapes by reasoning about similarities and differences in properties.	I can draw polygons by joining marked points, and identify parallel and perpendicular sides. I can make 3-D shapes using modelling materials; recognise 3-D shapes in different orientations and describe them.	I can identify regular polygons, including equilateral triangles and squares, as those in which the side-lengths are equal and the angles are equal. I can compare and classify geometric shapes, including quadrilaterals and triangles, based on their properties and sizes. I can identify line symmetry in 2D shapes presented in different orientations. I can reflect shapes in a line of symmetry and complete a symmetric figure or pattern with respect to a specified line of symmetry. I can identify acute and obtuse angles and compare and order angles up to two right angles by size.	I can compare angles, estimate and measure angles in degrees (°) and draw angles of a given size. I can identify 3-D shapes, including cubes and other cuboids, from 2-D representations. I can identify: angles at a point and one whole turn (total 360°); angles at a point on a straight line and $\frac{1}{2}$ a turn (total 180°); other multiples of 90°. I can use the properties of rectangles to deduce related facts and find missing lengths and angles. I can distinguish between regular and irregular polygons based on reasoning about equal sides and angles.	I can draw, compose, and decompose shapes according to given properties, including dimensions, angles and area, and solve related problems. I can compare and classify geometric shapes based on their properties and sizes and find unknown angles in any triangles, quadrilaterals, and regular polygons. I can illustrate and name parts of circles, including radius, diameter and circumference and know that the diameter is twice the radius. I can recognise angles where they meet at a point, are on a straight line, or are vertically opposite, and find missing angles.

Vocabulary		Circle, circular, corner, cube, cuboid, cylinder, 2-D; 3-D, kite, oblong, rectangle, side, square, triangle	Circle, circular, cone, corner, cube, cuboid, cylinder, 2-D; 3-D, hexagon, kite, oblong, octagon, pentagon, polygon, prism, pyramid, quadrilateral, rectangle, side, square, triangle, vertex	Circle, circular, composite shape, cone, corner, cube, cuboid, cylinder, 2-D; 3-D, edge, face, hexagon, kite, oblong, octagon, pentagon, polygon, prism, property, pyramid, quadrilateral, rectangle, side, square, triangle, vertex	Centre, composite shape, cone, cube, cuboid, curved surface, cylinder, 2-D; 3-D, diagonal, edge, face, hexagon, kite, oblong, octagon, parallel, parallelogram, pentagon, perpendicular, polygon, polyhedron, prism, property, pyramid, quadrilateral, rectangle, rectilinear, rhombus, sphere, square, tetrahedron, trapezium, triangle, vertex	Acute, angle, axis of symmetry, centre, composite shape, 2-D; 3-D, degree, diagonal, edge, face, hexagon, kite, line, oblong, obtuse, octagon, parallel, parallelogram, pentagon, perpendicular, polygon, polyhedron, prism, property, pyramid, quadrilateral, rectangle, rectilinear, reflection, regular, rhombus, sphere, square, symmetry, tetrahedron, trapezium, triangle, vertex	Acute, angle, angle at a point, angle at a point on a line, axis of symmetry, centre, composite shape, cone, cube, cuboid, curved surface, cylinder, 2-D; 3-D, degree, diagonal, dodecahedron, edge, face, heptagon, hexagon, icosahedron, line, net, obtuse, octahedron, parallel, parallelogram, pentagon, perpendicular, polygon, polyhedron, prism, property, protractor, pyramid, quadrilateral, rectangle, rectilinear, regular, rhombus, sphere, square, tetrahedron, trapezium, triangle, vertex	Acute, angle, angle at a point, angle at a point on a line, axis of symmetry, centre, circle, circular, circumference, compasses (pair of), composite shape, 2-D; 3-D, degree, diagonal, diameter, dodecahedron, edge, face, heptagon, hexagon, icosahedron, line, net, obtuse, octahedron, parallel, parallelogram, pentagon, perpendicular, polygon, polyhedron, prism, property, protractor, pyramid, quadrilateral, radius, rectangle, rectilinear, regular, rhombus, sphere, square, tetrahedron, trapezium, triangle, vertex, vertically opposite angles
------------	--	---	---	--	--	---	--	--

Geometry: Position & Direction

	Nursery (Mathematics)	Reception (Mathematics)	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can understand position through words alone, for example: "The bag is under the table," with no pointing.	I can select, rotate and manipulate shapes for a particular purpose, for example: rotating a cylinder so it can be used to build a tower, rotating a puzzle piece to fit in its place. I can use positional language to describe.	I can compose 2D and 3D shapes from smaller shapes to match an example, including manipulating shapes to place them in particular orientations. I can describe position, direction and movement including whole, half, quarter and three-quarter turns.	I can order and arrange combinations of mathematical objects in patterns and sequences. I can use mathematical vocabulary to describe position, direction and movement, including movement in a straight line and distinguishing between rotation as a turn and in terms of right angles for quarter, half and three-quarter turns (clockwise and anti-clockwise).	I can recognise right angles as a property of shape or a description of a turn, and identify right angles in 2D shapes presented in different orientations.	I can draw polygons, specified by coordinates in the first quadrant, and translate within the first quadrant.	I can identify, describe and represent the position of a shape following a reflection or translation, using the appropriate language, and know that the shape has not changed.	I can describe positions on the full coordinate grid (all four quadrants). I can draw and translate simple shapes on the coordinate plane, and reflect them in the axes.
Vocabulary	Under, above, between	Direction	Direction, quarter turn, rotation, turn	Angle, direction, quarter turn, right angle, rotation, turn	Angle, direction, right angle	Axis, Cartesian coordinate system, coordinate, direction, grid, horizontal, interval, origin, plot, position, quadrant, translation, vertical	Axis, Cartesian coordinate system, coordinate, direction, grid, horizontal, interval, origin, plot, position, quadrant, reflection, translation, vertical	Axis, Cartesian coordinate system, coordinate, direction, grid, horizontal, interval, origin, plot, position, quadrant, reflection, translation, vertical

Statistics					
	Year 2	Year 3	Year 4	Year 5	Year 6
Substantive knowledge & Skills	I can interpret and construct simple pictograms, tally charts, block diagrams and simple tables. I can ask and answer simple questions by counting the number of objects in each category and sorting the categories by quantity. I can ask and answer questions about totalling and comparing categorical data.	I can interpret and present data using bar charts, pictograms and tables. I can solve one-step and two-step questions [for example, 'How many more?' and 'How many fewer?'] using information presented in scaled bar charts and pictograms and tables.	I can interpret and present discrete and continuous data using appropriate graphical methods, including bar charts and time graphs. I can solve comparison, sum and difference problems using information presented in bar charts, pictograms, tables and other graphs.	I can solve comparison, sum and difference problems using information presented in a line graph. I can complete, read and interpret information in tables, including timetables.	I can interpret and construct pie charts and line graphs and use these to solve problems. I can calculate and interpret the mean as an average.
Vocabulary	Bar chart, Carroll diagram, chart, column graph, continuous data, Data, frequency, graph, pictogram, row, table, tally	Bar chart, chart, column, column graph, continuous data, data, frequency, graph, pictogram, row, table, tally	Bar chart, chart, column, column graph, continuous data, data, frequency, graph, pictogram, row, table	Chart, column, continuous data, data, frequency, graph, line graph, row, table	Average, chart, continuous data, data, frequency, graph, line graph, mean, pie-chart, row, table